# Module – 3

Sampling Design

Prepared by,
Jaswanth V,
Assistant Professor,
Dept of ECE, VVCE

Source: Research Methodology, CR Kothari

# Introduction - Field Survey

- 2 types
  - Census Survey
  - Sample Survey

# CENSUS AND SAMPLE SURVEY

- ## CENSUS SURVEY

- **Definition:** A census is a complete enumeration or count of all items in a defined population.

- **Explanation:**
  - In a census, every individual or element in the entire population is studied or surveyed.
  - The goal is to collect information from each member of the population, leaving no one out.
  - Census inquiries are presumed to provide the highest accuracy because they cover the entire population.
  - However, they can be resource-intensive, time-consuming, and may face challenges related to potential biases

- ## SAMPLE SURVEY:

- **Definition:** A sample is a subset of a population selected for study or analysis.

- **Explanation:**
  - In a sample survey, instead of studying the entire population, a representative subset (sample) is chosen.
  - The selection process involves using a predefined sampling technique to ensure that the sample is representative of the entire population.
  - Samples are more practical for large populations where a complete census may be impractical or too costly.
  - The goal is to draw conclusions about the entire population based on the characteristics observed in the selected sample.

# IMPLICATIONS OF A SAMPLE DESIGN

- **Sample Design:**
  - A plan to pick a smaller group (sample) from a larger group (population) for study.
  - Outlines how researchers will choose the sample and the number of items in it.
  - Decided before collecting data.
- **Determining Sample Size:**
  - Involves deciding how many items will be included in the sample.
  - Influences the precision and reliability of the study.
- **Timing of Sample Design:**
  - Plan is established before actual data collection begins.
  - Pre-emptive planning ensures a systematic approach to sampling.
- **Variety of Sample Designs:**
  - Different methods are available for researchers to choose from.
  - Some designs are more precise and easier to apply than others.
- **Importance of Reliability and Appropriateness:**
  - The chosen sample design must be both reliable and suitable for the study.
  - Reliability ensures consistent and dependable results; appropriateness aligns with research objectives.
- **Preceding Data Collection:**
  - Sample design decisions are made before starting to collect data.
  - Ensures a well-defined and systematic approach to the research study

- **Pre-emptive planning** - the term "preemptive" typically refers to actions or measures taken in advance to anticipate and address potential issues or challenges that may arise during the research process. It involves proactive steps to prevent or mitigate potential problems before they occur.

- **Reliability** in research design refers to the consistency, stability, or repeatability of measurements or findings. It is a crucial aspect of research quality, as reliable results are ones that can be consistently reproduced or replicated under similar conditions. If a measurement or instrument is reliable, it should produce consistent results when used repeatedly.

- The concept of **appropriateness** in research design pertains to the suitability and relevance of the chosen methods, procedures, and tools for addressing the research questions or objectives. A well-designed research study is not only rigorous but also appropriate in its approach, ensuring that the methods selected are the most fitting for the particular research context

# STEPS IN SAMPLE DESIGN – Type of Universe

1. **Type of Universe**
2. **Sampling Unit**
3. **Source List (Sampling Frame)**
4. **Size of Sample**
5. **Parameters of Interest**
6. **Budgetary Constraint**
7. **Sampling Procedure**

# STEPS IN SAMPLE DESIGN – Type of Universe

**Type of Universe:**

1. The first step in creating a sample design is to clearly define the set of objects under study, known as the "Universe."
2. The universe can be categorized as either finite or infinite, depending on the nature of the objects being studied.

**Finite Universe:**

In a finite universe, the number of items is certain and countable.
Examples of finite universes include the population of a city, the number of workers in a factory, and other situations where the total number of items is known and limited.

**Infinite Universe:**

1. In an infinite universe, the number of items is infinite, and there is no clear idea about the total number of items.
2. Examples of infinite universes include the number of stars in the sky, listeners of a specific radio program, and random events like throwing a dice.

Understanding whether the total number of items is known or unknown is crucial in determining the appropriate sampling method for the research study.

# STEPS IN SAMPLE DESIGN – Sampling Unit

**Definition of Sampling Unit:**

1. A sampling unit is a crucial decision made before selecting a sample for a study.

**Types of Sampling Units:**

1. The sampling unit can take various forms based on the nature of the study.
2. It may be a geographical unit, such as a state, district, or village.
3. Alternatively, it could be a construction unit, like a house or flat.
4. It might also be a social unit, such as a family, club, or school.
5. Finally, the sampling unit could be an individual.

**Decision on Sampling Unit:**

1. The researcher must decide which unit or units to select for the study.
2. Depending on the research objectives, one or more types of units may be chosen

## STEPS IN SAMPLE DESIGN – Source List

A sampling frame is essentially a list or a source that serves as the basis for selecting the sample for a research study. It includes all the elements or units that make up the population under study. The population could be finite or infinite, and the sampling frame should ideally cover all elements of the population.

Here are some key points about a sampling frame:

**1. Comprehensiveness:** The sampling frame should include all the elements of the population. If any elements are excluded, there's a risk of sampling bias, which may affect the generalizability of the study's findings.

**2. Correctness:** The information in the sampling frame should be accurate. Errors in the sampling frame can lead to inaccuracies in the sample selection and, consequently, affect the validity of the study.

**3. Reliability:** The sampling frame should be reliable, meaning that it should remain stable and consistent over time. Changes or fluctuations in the sampling frame can introduce variability and impact the study's reliability.

**4. Representativeness:** It is crucial for the sampling frame to be representative of the population. If the sampling frame is skewed or does not accurately reflect the characteristics of the entire population, the sample may not be truly representative.

**5. Appropriateness:** The sampling frame should be appropriate for the research question and objectives. It should align with the goals of the study and the characteristics of the population being studied.

In cases where an existing sampling frame is not available, researchers may need to create one

## STEPS IN SAMPLE DESIGN – Size of Sample

- This refers to the number of items to be selected from the universe to constitute a sample.
- This a major problem before a researcher.
- The size of sample should neither be excessively large, nor too small. It should be optimum.
- An optimum sample is one which fulfils the requirements of efficiency, representativeness, reliability and flexibility.
- While deciding the size of sample, researcher must determine the desired precision as also an acceptable confidence level for the estimate.
- The size of population variance needs to be considered -
    - it indicates that the data points are more spread out, and as a result, a larger sample size is often needed to obtain a representative sample that accurately reflects the characteristics of the entire population. This is because a larger sample size provides more information and helps to reduce the impact of variability, resulting in more reliable and precise estimates
- Costs too dictate the size of sample that we can draw. As such, budgetary constraint must invariably be taken into consideration when we decide the sample size.

## STEPS IN SAMPLE DESIGN – Parameters of Interest

- In determining the sample design, one must consider the question of the specific population parameters which are of interest.
- For instance, we may be interested in estimating the proportion of persons with some characteristic in the population, or we may be interested in knowing some average or the other measure concerning the population.
- There may also be important sub-groups in the population about whom we would like to make estimates.
- All this has a strong impact upon the sample design we would accept.

## STEPS IN SAMPLE DESIGN – Budgetary Constraint

- Cost considerations, from practical point of view, have a major impact upon decisions relating to not only the size of the sample but also to the type of sample. This fact can even lead to the use of a non-probability sample

**Note**: non-probability sample is a type of sampling method in which individuals are chosen for inclusion in a study based on non-random criteria, Convenience sampling (choosing elements which are easiest to reach/available, Purposive (choosing elements/individuals who are meeting certain criteria), Snowball (group of elements who recommends others to join) and Quota Sampling (choosing proportions considering all characteristics) are the types

## STEPS IN SAMPLE DESIGN – Sampling Procedure/Technique

- Finally, the researcher must decide the type of sample he will use i.e., he must decide about the technique to be used in selecting the items for the sample.
- In fact, this technique or procedure stands for the sample design itself.
- There are several sample designs, out of which the researcher must choose one for his study.
- Obviously, he must select that design which, for a given sample size and for a given cost, has a smaller sampling error.

## CHARACTERISTICS OF A GOOD SAMPLE DESIGN

From what has been stated above, we can list down the characteristics of a good sample design as under:

(a) Sample design must result in a truly representative sample.

(b) Sample design must be such which results in a small sampling error.

(c) Sample design must be viable in the context of funds available for the research study.

(d) Sample design must be such so that systematic bias (called as Systematic error) can be controlled in a better way. (Systematic error, also known as bias, refers to consistent and recurring errors that are introduced into the measurement process, leading to a deviation of results from the true value)

(e) When conducting research, it's crucial to have a sample that is representative of the larger population or universe from which it is drawn. This ensures that the findings from the sample study can be reasonably applied or generalized to the entire population.

DIFFERENT TYPES OF SAMPLE DESIGNS – two types based on the representation basis and the element selection technique

the representation basis, the sample may be **probability sampling** or it may be **non-probability sampling**

Probability sampling is based on the concept of random selection, whereas non-probability sampling is ‘non-random’ sampling

element selection technique – not in the syllabus

CHART SHOWING BASIC SAMPLING DESIGNS

| | Representation basis | |
|---|---|---|
| Element selection technique | Probability sampling | Non-probability sampling |
| Unrestricted sampling | Simple random sampling | Haphazard sampling or convenience sampling |
| Restricted sampling | Complex random sampling (such as cluster sampling, systematic sampling, stratified sampling etc.) | Purposive sampling (such as quota sampling, judgement sampling) |

## Non-probability sampling:

- Non-probability sampling is that sampling procedure which does not afford any basis for estimating the probability that each item in the population has of being included in the sample.

- Non-probability sampling is also known by different names such as deliberate sampling, purposive sampling and judgement sampling. In this type of sampling, items for the sample are selected deliberately by the researcher; his choice concerning the items remains supreme.

- In other words, under non-probability sampling the organisers of the inquiry purposively choose the particular units of the universe for constituting a sample on the basis that the small mass that they so select out of a huge one will be typical or representative of the whole.

- For instance, if economic conditions of people living in a state are to be studied, a few towns and villages may be purposively selected for intensive study on the principle that they can be representative of the entire state.

- Thus, the judgement of the organisers of the study plays an important part in this sampling design

## Non-probability sampling: (continue)

- In such a design, personal choice of element has a great chance of entering into the selection of the sample.
- The investigator may select a sample which shall yield results favourable to his point of view and if that happens, the entire inquiry may get vitiated (Corrupted/impaired). Thus, there is always the danger of bias entering into this type of sampling technique.
- But in the investigators are impartial, work without bias and have the necessary experience so as to take sound judgement, the results obtained from an analysis of deliberately selected sample may be tolerably reliable. However, in such a sampling, there is no assurance that every element has some specifiable chance of being included.
- Sampling error in this type of sampling cannot be estimated and the element of bias, great or small, is always there. As such this sampling design in rarely adopted in large inquires of importance. However, in small inquiries and researches by individuals, this design may be adopted because of the relative advantage of time and money inherent in this method of sampling.
- Quota sampling is also an example of non-probability sampling. Under quota sampling the interviewers are simply given quotas to be filled from the different strata, with some restrictions on how they are to be filled. In other words, the actual selection of the items for the sample is left to the interviewer's discretion. This type of sampling is very convenient and is relatively inexpensive. But the samples so selected certainly do not possess the characteristic of random samples. Quota samples are essentially judgement samples and inferences drawn on their basis are not amenable to statistical treatment in a formal way.

# Probability sampling:

- Probability sampling is also known as ‘random sampling’ or ‘chance sampling’.
- Under this sampling design, every item of the universe has an equal chance of inclusion in the sample. It is, so to say, a lottery method in which individual units are picked up from the whole group not deliberately but by some mechanical process.
- Here it is blind chance alone that determines whether one item or the other is selected. The results obtained from probability or random sampling can be assured in terms of probability i.e., we can measure the errors of estimation or the significance of results obtained from a random sample, and this fact brings out the superiority of random sampling design over the deliberate sampling design.
- Random sampling ensures the law of Statistical Regularity which states that if on an average the sample chosen is a random one, the sample will have the same composition and characteristics as the universe.
- This is the reason why random sampling is considered as the best technique of selecting a representative sample

## Probability sampling: (continue)

- Random sampling from a finite population refers to that method of sample selection which gives each possible sample combination an equal probability of being picked up and each item in the entire population to have an equal chance of being included in the sample.

- This applies to sampling without replacement i.e., once an item is selected for the sample, it cannot appear in the sample again (Sampling with replacement is used less frequently in which procedure the element selected for the sample is returned to the population before the next element is selected. In such a situation the same element could appear twice in the same sample before the second element is chosen).

- In brief, the implications of random sampling (or simple random sampling) are:

  - It gives each element in the population an equal probability of getting into the sample; and all choices are independent of one another.
  - It gives each possible sample combination an equal probability of being chosen

# Probability sampling(random) versus Deliberate Sampling(non-random): (An example)

**Scenario: Surveying Students in a School**

**Assurance in Terms of Probability (Point 1):**

- **Random Sampling:** Imagine you want to survey the opinions of students in a school with 500 students. If you use random sampling, each student has an equal chance of being selected.
- **Result:** Let's say you randomly select 50 students for your survey. Since it's random, any of the 500 students could have been chosen. This randomness assures that your sample is likely to represent the diversity of opinions in the entire school.

**Measuring Errors of Estimation (Point 2):**

- **Random Sampling:** After collecting responses, you find that 80% of the sampled students support a new school policy. You calculate a margin of error.
- **Result:** Your margin of error is ±5%. This means that you can be reasonably confident that if you had surveyed all 500 students, the true level of support for the policy in the entire school would likely fall within the range of 75% to 85%. The margin of error gives you a sense of how much your sample result might deviate from the actual population opinion

**Deliberate (Non-Random) Sampling:**

- **Selective Approach:** Instead of randomly selecting students, you decide to survey only students from the honor roll, assuming that their opinions might be more academically focused.
- **Result:** While this approach might provide insights from high-achieving students, it excludes the perspectives of other students who may not be on the honor roll. The sample is not representative of the entire student body.

**Challenges in Measuring Errors of Estimation:**

- **Limited Representation:** Since the sample is deliberately chosen from a specific group, it may not accurately represent the diversity of opinions among all students.
- **Result:** Without random sampling, it becomes difficult to measure the errors of estimation or determine how well the survey results reflect the opinions of the entire student population. The findings may be skewed toward the views of honor roll students.

# Measurement and Scaling Techniques - Introduction

- In our daily life we are said to measure when we use some yardstick to determine weight, height, or some other feature of a physical object.
- We also measure when we judge how well we like a song, a painting or the personalities of our friends.
- We, thus, measure physical objects as well as abstract concepts.
- Measurement is a relatively complex and demanding task, specially so when it concerns qualitative or abstract phenomena.
- By measurement we mean the process of assigning numbers to objects or observations, the level of measurement being a function of the rules under which the numbers are assigned.
- It is easy to assign numbers in respect of properties of some objects, but it is relatively difficult in respect of others.
- For instance, measuring such things as social conformity, intelligence, or marital adjustment is much less obvious and requires much closer attention than measuring physical weight, biological age or a person’s financial assets. In other words, properties like weight, height, etc., can be measured directly with some standard unit of measurement, but it is not that easy to measure properties like motivation to succeed, ability to stand stress and the like. We can expect high accuracy in measuring the length of pipe with a yard stick, but if the concept is abstract and the measurement tools are not standardized, we are less confident about the accuracy of the results of measurement

# Measurement and Scaling Techniques - Introduction

- Technically speaking, measurement is a process of mapping aspects of a domain onto other aspects of a range according to some rule of correspondence.

- In measuring, we devise some form of scale in the range (in terms of set theory, range may refer to some set) and then transform or map the properties of objects from the domain (in terms of set theory, domain may refer to some other set) onto this scale.

- For example, in case we are to find the male to female attendance ratio while conducting a study of persons who attend some show, then we may tabulate those who come to the show according to gender.

- In terms of set theory, this process is one of mapping the observed physical properties of those coming to the show (the domain) on to a gender classification (the range).

- The rule of correspondence is: If the object in the domain appears to be male, assign to “0” and if female assign to “1”.

# MEASUREMENT SCALES

- The scales of measurement can be considered in terms of their mathematical properties.

- The most widely used classification of measurement scales are:

(a) nominal scale;
(b) ordinal scale;
(c) interval scale; and
(d) ratio scale.

# MEASUREMENT SCALES

- **Nominal Scale**

  - Nominal scale is simply a system of assigning number symbols to events in order to label them.
  - The usual example of this is the assignment of numbers of basketball players in order to identify them.
  - Such numbers cannot be considered to be associated with an ordered scale for their order is of no consequence; the numbers are just convenient labels for the particular class of events and as such have no quantitative value.
  - Nominal scales provide convenient ways of keeping track of people, objects and events. One cannot do much with the numbers involved.
  - For example, one cannot usefully average the numbers on the back of a group of football players and come up with a meaningful value. Neither can one usefully compare the numbers assigned to one group with the numbers assigned to another.
  - The counting of members in each group is the only possible arithmetic operation when a nominal scale is employed. Accordingly, we are restricted to use mode as the measure of central tendency.

# MEASUREMENT SCALES

- **Nominal Scale**

  - Nominal scale is the least powerful level of measurement.

  - It indicates no order or distance relationship and has no arithmetic origin.

  - A nominal scale simply describes differences between things by assigning them to categories. Nominal data are, thus, counted data.

  - The scale wastes any information that we may have about varying degrees of attitude, skills, understandings, etc. In spite of all this, nominal scales are still very useful and are widely used in surveys and other ex-post-facto research when data are being classified by major sub-groups of the population

    **Example:** Colors, gender, types of fruits, or car brands are examples of nominal data.

- Used to label or categorize things, but cannot perform meaningful mathematical operations like addition or subtraction with nominal data.

# MEASUREMENT SCALES

- **Ordinal Scale – Strongly disagree to disagree to netural to agree to strongly agree, difference between them is not the concern, ordering only**
  - The ordinal scale places events in order, but there is no attempt to make the intervals of the scale equal in terms of some rule.
  - Rank orders represent ordinal scales and are frequently used in research relating to qualitative phenomena.
  - A student's rank in his graduation class involves the use of an ordinal scale.
  - One has to be very careful in making statement about scores based on ordinal scales. For instance, if Ram's position in his class is 10 and Mohan's position is 40, it cannot be said that Ram's position is four times as good as that of Mohan.
  - The statement would make no sense at all.
  - Ordinal scales only permit the ranking of items from highest to lowest.
  - Ordinal measures have no absolute values, and the real differences between adjacent ranks may not be equal. All that can be said is that one person is higher or lower on the scale than another, but more precise comparisons cannot be made.

# MEASUREMENT SCALES

- **Ordinal Scale**
  - Thus, the use of an ordinal scale implies a statement of ‘greater than’ or ‘less than’ (an equality statement is also acceptable) without our being able to state how much greater or less.
  - The real difference between ranks 1 and 2 may be more or less than the difference between ranks 5 and 6.

# MEASUREMENT SCALES

- **Interval Scale**
  - In the case of interval scale, the intervals are adjusted in terms of some rule that has been established as a basis for making the units equal.
  - The units are equal only in so far as one accepts the assumptions on which the rule is based. Interval scales can have an arbitrary zero, but it is not possible to determine for them what may be called an absolute zero or the unique origin.
  - The primary limitation of the interval scale is the lack of a true zero; it does not have the capacity to measure the complete absence of a trait or characteristic.
  - The Fahrenheit scale is an example of an interval scale and shows similarities in what one can and cannot do with it.
  - One can say that an increase in temperature from 30° to 40° involves the same increase in temperature as an increase from 60° to 70°, but one cannot say that the temperature of 60° is twice as warm as the temperature of 30° because both numbers are dependent on the fact that the zero on the scale is set arbitrarily at the temperature of the freezing point of water.
  - The ratio of the two temperatures, 30° and 60°, means nothing because zero is an arbitrary point.

# MEASUREMENT SCALES

- **Interval Scale**

- Interval scales provide more powerful measurement than ordinal scales for interval scale also incorporates the concept of equality of interval

**Note**

**Definition:** In an interval scale, data points have equal intervals or distances between them, and zero does not represent the absence of the quantity being measured.

• **Example:** Temperature measured in Celsius or Fahrenheit is an interval scale. The difference between 20 and 30 degrees is the same as the difference between 30 and 40 degrees.

•**Key Points:**

- Equal intervals between points.
- Arbitrary zero point (zero does not mean the absence of the measured quantity).
- Allows for meaningful comparison of differences between values.

In summary, an interval scale goes beyond just naming or categorizing; it provides information about the relative distances between different points on the scale. However, it's important to note that an interval scale doesn't have a true zero point, and ratios (like a temperature of 60 degrees being twice as hot as 30 degrees) are not meaningful.

# MEASUREMENT SCALES

- **Ratio Scale**

- Ratio scales have an absolute or true zero of measurement.

A ratio scale is a type of measurement scale that possesses all the properties of an interval scale but with the additional feature of having a true zero point. Here are the key characteristics of ratio scales:

**Equal Intervals:** Like an interval scale, a ratio scale has equal intervals between consecutive points on the scale. The differences between values are meaningful and consistent.

**True Zero Point:** The crucial feature that distinguishes a ratio scale is the presence of a true zero point. A true zero point means that zero on the scale indicates the complete absence of the quantity being measured, and it is not merely an arbitrary or starting point.

**Meaningful Ratios:** With a true zero point, meaningful ratios can be calculated. For example, if one value is twice as large as another, it indicates a true twofold difference in the quantity being measured.
**Mathematical Operations:** In addition to meaningful ratios, mathematical operations like multiplication and division are meaningful on a ratio scale.

Examples of variables measured on a ratio scale include height, weight, income, age (when measured in years), distance, and time (when measured in seconds). These variables not only have equal intervals between values but also have a true zero point, allowing for a more extensive range of meaningful statistical analyses and interpretations

# Sources of Error in Measurement

- **Respondent:**
  - Reluctance
  - Transient factors
  - Fatigue
  - Boredom
  - Anxiety
- **Situation:**
  - Presence of others
  - Anonymity concerns
  - Strain on rapport
- **Measurer:**
  - Rewording
  - Reordering questions
  - Behavior
  - Style
  - Looks
  - Mechanical errors
  - Incorrect coding
  - Faulty tabulation
  - Statistical errors
- **Instrument:**
  - Defective measuring instrument
  - Complex words
  - Ambiguous meanings
  - Poor printing
  - Inadequate space
  - Response choice omissions
  - Poor sampling
- **Researcher's Responsibilities:**
  1. Eliminate
  2. Neutralize
  3. Address sources of error
  4. Design
  5. Training
  6. Consistency

**Note:** Researchers use various methods to collect information from respondents, such as surveys, interviews, focus groups, or observational studies. The responses provided by respondents contribute to the data that researchers analyze to draw conclusions, make inferences, or gain insights into specific topics or issues.

# Sources of Error in Measurement

**1.Respondent:**

1. *Reluctance to Express Feelings:* A respondent might be hesitant to admit ignorance or express strong negative feelings, leading to inaccurate responses. For example, a participant might be unwilling to admit that they don't understand a question.
2. *Transient Factors:* Fatigue, boredom, or anxiety can affect a respondent's ability to provide accurate information. For instance, a fatigued participant may provide hasty and less thoughtful responses.

**2.Situation:**

1. *Presence of Others:* If someone else is present during an interview, it can influence responses. A participant might alter their answers to conform to social expectations or might feel reluctant to express certain feelings in the presence of others.
2. *Anonymity Concerns:* If respondents feel that their anonymity is not assured, they may withhold sensitive information, leading to biased or incomplete responses.

**3.Measurer(Interviewer):**

1. *Rewording or Reordering Questions:* The way a question is phrased can impact responses. If a question is reworded or reordered, it may introduce confusion or bias. For example, asking "Do you support policy X?" may yield different responses than asking "What are your thoughts on policy X?"
2. *Behavior, Style, Looks:* The demeanor of the interviewer, their body language, or appearance can influence responses. A participant might provide socially desirable answers based on the interviewer's behavior.

Note: Respondent factors can impact the quality of responses, situation factors can influence how comfortable respondents fee
and measurer factors relate to how the survey is designed and administered.

# Sources of Error in Measurement

**3) Instrument:**

- *Defective Measuring Instrument:* If the survey or measurement tool is flawed, it can introduce errors. For instance, using complex words or ambiguous terms in a questionnaire may lead to misinterpretation by respondents.
- *Poor Sampling:* If the instrument doesn't adequately represent the full range of items of concern, the results may lack generalizability – (Generalizability is the extent to which research findings can be applied or generalized to a larger population beyond the study sample. If the sample or instrument is not representative, the results may not be applicable to the broader context).

**4)Researcher's Responsibilities:**

- *Eliminate, Neutralize, or Address Sources of Error:* Researchers must be proactive in minimizing these errors. For example, they can design surveys with clear and simple language, ensure participant privacy, and provide detailed training to interviewers to maintain consistency

In summary, understanding and addressing these factors are crucial for a researcher to enhance the validity and reliability of their study's measurements. It emphasizes the importance of careful planning, thoughtful design, and meticulous execution in the research process.

# Test of Sound Measurement

Sound measurement must meet the tests of validity, reliability and practicality

- **Definition: Validity** refers to the extent to which a test measures what it is intended to measure.
- **Significance:** A valid measurement tool accurately captures the specific construct or attribute it aims to assess. For sound measurement, this would mean that the tool effectively measures the intended aspects of sound, such as volume, frequency, or quality.
- **Example:** If a sound meter is designed to measure decibel levels, it would be considered valid if it accurately reflects the true decibel levels of the sound in question.

- **Definition: Reliability** is concerned with the accuracy and precision of a measurement procedure.
- **Significance:** A reliable measurement tool produces consistent and stable results over repeated measurements under the same conditions. In the context of sound measurement, a reliable tool would provide consistent results when measuring the same sound multiple times.
- **Example:** If a sound meter consistently provides similar decibel readings for the same sound source in repeated trials, it can be considered reliable.

- **Definition: Practicality** is concerned with factors such as economy, convenience, and interpretability of a measurement tool.
- **Significance:** A practical measurement tool is one that is efficient, cost-effective, easy to use, and provides results that are easy to interpret. This ensures that the measurement process is feasible and applicable in real-world situations.
- **Example:** A practical sound measurement tool might be portable, user-friendly, and not overly expensive, making it accessible for a wide range of applications.

Test of Validity - three types of **validity**

(i) Content validity; (ii) Criterion-related validity and (iii) Construct validity

- **Content validity** is the extent to which a measuring instrument provides adequate coverage of the topic under study.
- If the instrument contains a representative sample of the universe, the content validity is good.
- Its determination is primarily judgmental and intuitive.
- It can also be determined by using a panel of persons who shall judge how well the measuring instrument meets the standards, but there is no numerical way to express it

- **Criterion-related validity** relates to our ability to predict some outcome or estimate the existence of some current condition
- The concerned criterion must possess the following qualities,
    - Relevance
    - Freedom from bias
    - Reliability
    - Availability

- CR also referred to (i) Predictive validity - refers to the usefulness of a test in predicting some future performance.
- (ii) Concurrent validity - refers to the usefulness of a test in closely relating to other measures of known validity.

- (iii) Construct validity - A measure is said to possess construct validity to the degree that it confirms to predicted correlations with other theoretical propositions

## Test of Reliability

- The test of reliability is another important test of sound measurement.
- A measuring instrument is reliable if it provides consistent results.
- Reliable measuring instrument does contribute to validity, but a reliable instrument need not be a valid instrument.
- For instance, a scale that consistently overweighs objects by five kgs., is a reliable scale, but it does not give a valid measure of weight.

- Two aspects of reliability viz., **stability and equivalence**.

- The **stability** aspect is concerned with securing consistent results with repeated measurements of the same person and with the same instrument. We usually determine the degree of stability by comparing the results of repeated measurements.

- The **equivalence** aspect considers how much error may get introduced by different investigators or different samples of the items being studied

## Test of Practicality

the measuring instrument ought to be practical i.e., it should be **economical, convenient and interpretable.**

**Economical -**

- trade-off is needed between the ideal research project and that which the budget can afford
- length of measuring instrument is an important area where economic pressures are quickly felt
- in the interest of limiting the interview or observation time, we have to take only few items for our study purpose
- data-collection methods to be used are also dependent at times upon economic factors

**Convenience -**

- test suggests that the measuring instrument should be easy to administer.
- one should give due attention to the proper layout of the measuring instrument
- For instance, a questionnaire, with clear instructions (illustrated by examples), is certainly more effective and easier to complete than one which lacks these features.

**Interpretability** to interpret the results.

- The measuring instrument, in order to be interpretable, must be supplemented by
- a) detailed instructions for administering the test;
- (b) scoring keys;
- (c) evidence about the reliability and (d) guides for using the test and for interpreting results.

# Module - 4

- Methods of Data Collection
  - COLLECTION OF PRIMARY DATA – based on Experimental (data collected during performing experiments) research, descriptive research(to perform surveys by observation etc)

**(i) observation method, (ii) interview method**, (iii) through questionnaires, (iv) through schedules, and (v) other methods which include (a) warranty cards; (b) distributor audits; (c) pantry audits; (d) consumer panels; (e) using mechanical devices; (f) through projective techniques; (g) depth interviews, and (h) content analysis.

## Observation method

- behavioural sciences.

- observation is not scientific observation. Observation becomes a scientific tool and the method of data collection
- the information is sought by way of investigator’s own direct observation without asking from the respondent

- The main advantage of this method is that subjective bias is eliminated, if observation is done accurately. Secondly, the information obtained under this method relates to what is currently happening; it is not complicated by either the past behaviour or future intentions or attitudes. Thirdly, this method is independent of respondents’ willingness to respond and as such is relatively less demanding of active cooperation

- particularly suitable in studies which deal with subjects (i.e., respondents) who are not capable of giving verbal reports of their feelings for one reason or the other

- Observation method has various limitations. Firstly, it is an expensive method. Secondly, the information provided by this method is very limited. Thirdly, sometimes unforeseen factors may interfere with the observational task

**Interview method**

- The interview method of collecting data involves presentation of oral-verbal stimuli and reply interms of oral-verbal responses.

  *1. Personal interviews* requires a person known as the interviewer asking questions generally in a face-to-face contact to the other person or in groups, **direct personal investigation or it may be indirect oral investigation**, structured and non-structured interviews, focussed interview – focus on one thing like on **given** experience of the respondent, clinical interview – to talk on respondent's life experience **(not given)** and the non-directive interview – to talk about a certain topic

  *2. Telephone interviews* - This method of collecting information consists in contacting respondents on telephone itself.
  Merits: It is not a very widely used method, It is more flexible in comparison to mailing method, It is faster than other methods i.e., a quick way of obtaining information, It is cheaper than personal interviewing method; here the cost per response is relatively low, Recall is easy; callbacks are simple and economical.
  Demerits: Little time is given to respondents for considered answers; interview period is not likely to exceed five minutes in most cases. Surveys are restricted to respondents who have telephone facilities. Extensive geographical coverage may get restricted by cost considerations

## COLLECTION OF DATA THROUGH QUESTIONNAIRES

- This method of data collection is quite popular, particularly in case of **big enquiries**. It is being adopted by private individuals, research workers, private and public organisations and even by governments.

- In this method a questionnaire is sent (usually by post) to the persons concerned with a request to answer the questions and return the questionnaire. A questionnaire consists of a number of questions printed or typed in a definite order on a form or set of forms. The questionnaire is mailed to respondents who are expected to read and understand the questions and write down the reply in the space meant for the purpose in the questionnaire itself. The respondents have to answer the questions on their own.

**Merits** There is low cost even when the universe is large and is widely spread geographically, It is free from the bias of the interviewer; answers are in respondents' own words, Respondents have adequate time to give well thought out answers, Respondents, who are not easily approachable, can also be reached conveniently

**Demerits** It can be used only when respondents are educated and cooperating. The control over questionnaire may be lost once it is sent. There is also the possibility of ambiguous replies or omission of replies altogether to certain questions; interpretation of omissions is difficult. It is difficult to know whether willing respondents are truly representative

Quite often questionnaire is considered as the heart of a survey operation. Ex: Pilot Survey

Types/main aspects of a questionnaire:

- *General form – Structured and Unstructured*
- *Question sequence – start with introduction, purpose of the questionnaire, warm-up questions, main question, closing questions etc*
- *Question formulation and wording* – should be easily understood; should be simple

**1.Structured Questionnaires**: These questionnaires consist of predetermined, concrete questions presented in a standardized format. All respondents receive the same set of questions, typically with closed-ended (yes/no) or fixed-alternative questions. The goal is to ensure consistency in responses and ease of analysis.

**2.Unstructured Questionnaires**: In contrast, unstructured questionnaires provide a general guide for the interviewer, but the exact question formulation is left to the interviewer's discretion. Responses are typically recorded in the respondent's own words, allowing for greater flexibility and depth of information. Tape recorders may be used to capture responses accurately.

## Interpretation and Report Writing

- After collecting and analyzing the data, the researcher has to accomplish the task of **drawing inferences followed by report writing**
- It is only through interpretation that the researcher can expose relations and processes that underlie his findings.
- All this analytical information and consequential inference(s) may well be communicated, preferably through research report

The task of interpretation has two major aspects viz., (i) the **effort to establish continuity in research** through linking the results of a given study with those of another, and (ii) the **establishment of some explanatory concepts.**

## TECHNIQUE OF INTERPRETATION

- The task of interpretation is not an easy job, rather it requires a great skill and dexterity on the part of researcher. Interpretation is an art that one learns through practice and experience.

- Some techniques:
  In research, the researcher should explain findings clearly, interpret relationships, and look for common patterns across different results.
- Extra information collected during the study should be taken into account when interpreting the final results, as it could be important for understanding the problem being studied

- Before final interpretation, it's wise to seek input from someone knowledgeable about the study, who is honest and willing to point out any omissions or errors, ensuring accurate interpretation and enhancing the usefulness of research results.

- Before interpreting results, researchers should carefully consider all relevant factors affecting the problem to prevent false generalizations. Rushing through interpretation can lead to inaccurate conclusions, as initial impressions may not always be correct.

**DIFFERENT STEPS IN WRITING REPORT**

1. **Logical analysis of the subject-matter**: Understand the topic thoroughly and identify key points and relevant information.
2. **Preparation of the final outline**: Organize the information logically by creating a structured outline that outlines the main sections and sub-sections of the report.
3. **Preparation of the rough draft**: Write the initial version of the report based on the outline, focusing on getting ideas down without worrying too much about grammar or style.
4. **Rewriting and polishing**: Revise and refine the rough draft, improving clarity, coherence, and readability, and ensuring that the report flows smoothly.
5. **Preparation of the final bibliography**: Compile a list of all sources cited in the report, following a specific citation style (e.g., APA, MLA), and ensure that all sources are properly referenced.
6. **Writing the final draft**: Incorporate any final revisions and edits, proofread the report for errors, and produce the final version ready for submission or publication.

**APA (American Psychological Association)   MLA (Modern Language Association)**

## LAYOUT OF THE RESEARCH REPORT

- The layout of the report means as to what the research report should contain.
- A comprehensive layout of the research report should comprise (A) preliminary pages; (B) the main text; and (C) the end matter

**Preliminary pages** - *title and date*, acknowledgements in the form of ‘Preface’ or ‘Foreword’, *table of contents* followed by *list of tables and illustrations*

**Main text**: (i) Introduction; (ii) Statement of findings and recommendations; (iii) The results; (iv) The implications drawn from the results; and (v) The summary.

**End Matter**: questionnaires, sample information, mathematical derivations and the like ones. Bibliography of source consulted should also be given. Index (an alphabetical listing of names, places and topics along with the numbers of the pages in a book or report on which they are mentioned or discussed)

**MECHANICS OF WRITING A RESEARCH REPORT**

1. **Manuscript specifications**: Use specific paper size and ink color, with adequate margins and legible writing.
2. **Procedure adherence**: Follow the outlined steps rigorously as discussed previously.
3. **Layout planning**: Tailor the report layout to the problem's nature and objectives.
4. **Quotation handling**: Incorporate quotations appropriately, adjusting spacing and indentation for longer quotes.
5. **Footnotes management**: Number and format footnotes correctly, providing supplemental information without overusing them.
6. **Documentation style**: Properly format references, including author names, titles, publication details, and page numbers.
7. **Punctuation and abbreviation**: Use common abbreviations and punctuations in footnotes for clarity and conciseness.
8. **Statistical presentation**: Use statistics, charts, and graphs effectively to illustrate and simplify research findings.
9. **Final draft preparation**: Carefully revise and rewrite the rough draft to ensure clarity, logic, and coherence.
10. **Bibliography inclusion**: Compile and append a bibliography to the report.
11. **Index creation**: Prepare an alphabetical index, both subject-based and author-based, to aid readers in navigating the report effectively.

**Manuscript – a article, paper document before its published**